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Reassessment of sentence combining and decombining: The effects of sentence manipulation activities on reading comprehension

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Abstract

Purpose – The purpose of this paper is to examine the effects of sentence combining (SC) and sentence decombining (SD) activities on fostering reading comprehension. As a widely used writing activity for enhancing syntactic fluency in English Language Arts (ELA) classes, SC requires learners to combine short sentences into longer and more complex sentences, while SD requires learners to break down a long sentence into the shortest grammatically allowable sentences.

Design/methodology/approach – This study assessed the effects of SD and SC in comparison with a control group on the improvement of reading comprehension ability among college students learning English as their second language (L2) in the context of a six-week English language learning program. Participants with overall intermediate English language proficiency were randomly assigned to one of three different conditions: SC, SD and control. Also, a subset of the participants was interviewed after the intervention.

Findings – The results showed that SD was more effective than SC or control condition in enhancing syntactic knowledge and reading comprehension, as measured by a standardized English proficiency test. Data obtained from post-study interviews further suggested that only SD was perceived by the participants as having enhanced their reading comprehension.

Originality/value – The present study provides a valuable addition to a body of research on sentence manipulation activities in ELA classes. For those L2 learners who have passed a pre-intermediate threshold level, SD appears to be more beneficial than SC in enhancing syntactic knowledge, which, in turn, appears to contribute to better reading comprehension.

Keywords L2 learners, Reading comprehension, Sentence combining, Sentence decombining, Syntactic knowledge

Paper type Research paper

Introduction

Sentence combining (SC) is a widely used instructional technique for the development of writing skills (Graham and Perin, 2007). In this approach, students are given several short sentences and asked to combine them into a longer and more complex sentence by applying syntactic knowledge. Early studies of SC (Mellon, 1969; O'Hare, 1973) documented the positive effects of this instructional approach on the

development of syntactic fluency in composition among lower-secondary students in English Language Arts (ELA) classes, and since then, SC has attracted studies concentrating on the same issues with other age-group learners such as students in secondary-and-above levels (Daiker *et al.*, 1978; Keen, 2004). Moreover, research on SC has expanded the technique's potential for fostering reading comprehension (Graham and Hebert, 2010). Along these lines, a number of studies have attempted to demonstrate the effects of SC on reading comprehension, in particular with upper elementary students (Neville and Searls, 1985; Wilkinson and Patty, 1993).

Sentence decombining (SD), the flip side of the conceptual coin of SC, has also been considered as a type of effective writing activity by some researchers (Ney, 1976; Strong, 1986). In SD activities, students are required to reverse what one usually does in SC exercises, and to break down a long, complex sentence into shorter ones. However, unlike SC, SD has not received much attention in the field. Thus, there have been a limited number of opportunities to verify the effectiveness of SD for syntactic fluency or reading comprehension as an instructional method.

Taken together, the aim of the present study is to examine the effects of SC and SD activities, in comparison with a control condition, on fostering reading comprehension among undergraduate students learning English as a second language (L2) in the context of a six-week English learning program. With the lack of a previous study considering the SC and SD effects for L2 learners, the findings of the present study shall have pedagogical implications for many ELA classrooms in English-speaking countries (Australia, the UK, the USA, etc.) in light of internationalization of student populations in these countries (Altbach and Knight, 2007), where both native English speakers (NESs) and English language learners (ELLs) learn academic literacy. In the following section, the authors examine theoretical links among sentence manipulation activities, syntactic knowledge (awareness) and reading comprehension, as suggested by previous research (Fitzgerald and Shanahan, 2000; Stotsky, 1975; Straw and Schreiner, 1982). It should be noted here that English is the target language both for the first language (L1) and L2 contexts discussed in the current article.

Background

Reading research has focused on different dimensions or aspects that may contribute to reading comprehension: for example, from lower-level processing, such as word recognition, orthographic processing and phonological processing (Cunningham *et al.*, 2001; Holmes, 2009), to higher-level processing based on syntactic knowledge (Nation and Snowling, 2000). Among these dimensions, the importance of syntactic knowledge has been strongly highlighted not only in L1 studies (Cain, 2007; Nation and Snowling, 2000) but also in L2 studies (Bernhardt, 1991; Shiotsu and Weir, 2007). As the importance of syntactic knowledge in reading comprehension becomes ever clearer (Alderson, 2000; Bernhardt, 1991), continuous efforts have been made to enhance readers' syntactic knowledge. SC, which will be discussed next, is one of the explored approaches in this regard.

Sentence combining in writing and reading research

Early researchers examined the effects of SC exercises on the development of syntactic complexity in the writing of lower-secondary students (e.g. seventh grade in Mellon, 1969; O'Hare, 1973) in L1 contexts. Strong (1986, p. 5) describes SC exercises as "language play", through which learners are asked to merge a

few simple or short sentences into a longer one. Strong also notes that such exercises have the potential to enhance syntactic knowledge or awareness on the part of learners. Along these lines, researchers have validated the positive effects of SC on enhancing writing skills of students in secondary-and-above levels in L1 contexts (Daiker *et al.*, 1978; Keen, 2004) by demonstrating the development of students' syntactic maturity in writing through SC exercises. As a result, SC has risen as an alternative to "the teaching of formal sentence grammar" in enhancing writing quality (Andrews, 2005, p. 70).

Stotsky (1975), on the other hand, proposed the potential of SC to enhance reading ability. Given that SC is beneficial to the development of syntactic knowledge, she suggested that SC would contribute to enhancing other components of language. Similarly, Straw and Schreiner (1982) proposed a possible link between sentence manipulation exercises (which are, by nature, writing-oriented), syntactic knowledge and other skills, such as listening and reading, positing that syntactic knowledge is a common ground between reading and writing skills (Fitzgerald and Shanahan, 2000). A good number of empirical findings (Neville and Searls, 1985; Wilkinson and Patty, 1993) are in line with this hypothesis. For example, Wilkinson and Patty gave fourth-graders 16 sessions of SC activities, using three testing instruments (i.e. the standardized reading comprehension test, content-word-based test and structure/function-word-based cloze tests). The results showed that SC had positive effects on reading comprehension and cohesion knowledge (as measured by structure/function-word deleted cloze tests), when compared with the effects of a placebo treatment for the control group. Thus, this study offers an important piece of evidence on the link among SC, syntactic knowledge and reading comprehension.

SC, however, has received much less attention in L2 research, and findings from related studies are sometimes at odds with one another. Among a handful of studies, Klassen's study (1977, cited in Enginarlar, 1994) showed that systematic SC activities may result in higher syntactic maturity in L2 writers. On the other hand, Perkins and Konneker's (1982) study, which looked into L2 college students registered in a freshman composition course, showed that SC did not result in better outcomes in terms of their writing skills than did traditional instruction. Overall, the lack of studies specifically on L2 students leaves the value of SC in enhancing L2 learners' syntactic knowledge or awareness uncertain, let alone its effect on reading comprehension.

Sentence decombining in writing and reading research

Although the effects of SC on writing and reading have been examined to some extent, fewer studies have been dedicated to SD in L1 contexts (and our search has failed to find any SD studies in L2 contexts). Apart from Strong (1986), who hinted that SD may contribute to the enhancement of syntactic knowledge among advanced learners, only a limited number of authors (Neville and Searls, 1985; Ney, 1976) have discussed its effects on reading and writing, and their findings are somewhat unclear. For example, Ney (1976) tested the effects of sentence decomposition-recomposition (which can be thought of as a combination of SD and SC) and SC on the syntactic structure of L1 undergraduate freshmen's compositions. The results revealed that students in the sentence decomposition-recomposition group used a lower number of practiced syntactic structures in their compositions than those did in the SC group. These results coincided with those of the follow-up survey, in which one-third of the decomposition–recomposition group reported that they considered composition exercises to not be particularly useful.

With sixth graders in L1 contexts, Neville and Searls (1985) examined the combined effects of kernel-identification exercises (which is roughly equivalent to SD) and SC activities, in comparison with those of cloze test practices (the control condition), using cloze tests and questions taken from standardized reading comprehension tests. They found that the experimental group outperformed the control group on cloze tests, but not on the reading comprehension tests. However, these findings should be interpreted with caution, as it is not clear whether the outcomes resulted from kernel-identification training or SC activities.

The rationale of the present study

In light of the literature review presented above, which identified a general lack of L2 studies of SC, unclear findings on the effects of SD in L1 contexts, there appears to be a serious gap in the body of research on sentence manipulation activities. To close this gap, the present study addresses the following two issues: first, the question of whether SD may enhance knowledge or awareness of syntactic structures, as well as reading comprehension, has not received the attention it deserves. Second, only a small number of studies (Perkins and Konneker, 1982) have explored the use of SC in L2 contexts. To this end, the present study examines the effects of SD and SC on reading comprehension in a group of undergraduate L2 learners with an intermediate level of English language proficiency. The following three research questions are addressed:

RQ1. Are there any significant differences among different groups in their performance on the syntactic section of a standardized English proficiency test?

RQ2. Are there any significant differences among different groups in their performance on the reading comprehension section of a standardized English proficiency test?

RQ3. How do the participants perceive the value of sentence manipulation activities?

Methods

Participants and context

In total, 45 South Korean L2 undergraduates aged 19 to 22 years participated in the present study. All participants had studied English in formal contexts for about nine years during the primary and secondary education, and had taken two mandatory “English Reading” courses during the first year of their undergraduate study. By the time of the study, the participants were registered in a 6-week English learning program that aimed to enhance four language skills. The average score of the participants on the Test of English for International Communication (TOEIC) prior to the program was 645, indicating that they were beyond the threshold level (B1) based on the Common European Framework of Reference for Languages (CEFR) (Educational Testing Service, 2013b). The impetus for targeting L2 learners of approximately this proficiency comes from Zamel’s (1980) prediction that one of the reasons for the paucity of L2 studies on sentence manipulation activities may be that L2 learners generally do not have “the linguistic repertoire that sentence-combining proponents assume students have” (p. 84). In light of the findings of Zamel (1980), Strong (1986), and Bernhardt (1991), L2 learners of approximately this proficiency may be at a better stage for absorbing syntactic knowledge from this type of instruction, which in turn would lead to an improvement in reading comprehension. The participants were randomly assigned to three different groups: an SD group (n = 15), an SC group (n = 15), and a control group (n = 15). The results of analyses of variance (ANOVAs) for the pre-test (details are given below) confirmed the homogeneity between groups (p

> 0.05). Apart from the treatments provided in the reading classes, the three groups all followed the same curriculum and used the same materials during the program. Each group was given lessons in different classes.

Testing instrument

Two different sets of an official TOEIC mock test in a parallel form (Educational Testing Service, 2009, 2013a) were administered to all participants, with one test given at the beginning of the program (i.e. the pre-test; before the piloting) and the other at the end (i.e. the post-test). The authors selected this particular test because one of the objectives of this program was to help students augment their scores on the TOEIC. To measure the effects of the treatments, the participants' scores on test items directly related to syntactic knowledge and reading comprehension were obtained. The reading section of the TOEIC consists of a total of 100 multiple-choice questions. As each question is worth five points, the maximum score for the reading section is 500. The reading section is subdivided into three parts in terms of the length of the prompt (i.e. Parts 5, 6, and 7; Parts 1 through 4 are subsections of the listening section): (a) Part 5 is a cloze type test, in which one sentence is given for each question; (b) Part 6 is also a cloze type test, but a short text (usually more than one paragraph) is given for three to four questions; and (c) Part 7 is a reading comprehension test, in which a text of varying length is given, along with two to four questions on the target text. Parts 5 and 6 mainly focus on evaluating syntactic (grammatical) knowledge and vocabulary, and the last part emphasizes reading comprehension (Educational Testing Service, 2009, 2013a). Through a careful examination of the testing items in the pre- and post-tests used in the present study, the authors found that the aforementioned Parts 5, 6 and 7 of the reading section consisted of 21 questions that assess vocabulary, 31 questions that assess syntactic (grammatical) knowledge, and 48 questions that assess reading comprehension. Sample questions publicized by the developer (i.e. Educational Testing Service) are illustrated in Appendix 1.

Piloting

The first week of the six-week program was dedicated to piloting and optimizing the target treatments to the participants. One instructor, who had three years of experience teaching English at the tertiary level, was in charge of the reading lessons for all three groups. The teaching methods and materials were the same across groups, thereby ensuring that the only difference was the treatment designed for each condition. The purpose of the reading lessons was to enhance reading comprehension skills by exposing the students to a wide range of passages excerpted from a TOEIC textbook (Cho, 2008).

Three 70-min reading lessons were given during the first week. The authors were present for the sake of treatment fidelity. Each group received its respective treatment for 30 min: first, participants in the SD group were asked to decombine a set of sentences that were extracted from the target reading passage into two or three simple sentences. Second, for the SC group, the authors decombined sentences from the target reading passage prior to each class. The participants in the SC group were asked to combine these short sentences into a single longer sentence. Third, participants in the control group received the same reading passage as the other two groups. However, participants in the control group were given more time to thoroughly read the passage, as no instruction was offered to this group. After receiving the treatments, all participants were asked to identify the topic sentence or main ideas in the target reading passage, and to complete the reading comprehension questions.

For the purpose of piloting, the authors tried to focus on which types of syntactic structures the participants found challenging when they were reading different kinds of passages included in the aforementioned TOEIC textbook. After the piloting, the authors and the instructor were able to identify the syntactic structures that were particularly challenging for the participants across conditions. These were then used to draw up the list of syntactic structures that were to be dealt with in the main study (see Appendix 2 for this list).

During the pilot period, the authors found that a majority of participants did not apply the SD or SC activities as they originally intended. For example, some participants in the SC group combined short sentences simply by using connectors, such as “and” and “but”, regardless of target syntactic structures. Participants in the SD group varied widely in terms of the number of simple sentences that they made or syntactic structures that they identified in breaking down a long sentence.

Thus, the modification was made to the original SD and SC instructional techniques. That is, the authors included additional cues for each target sentence (Strong, 1986; Wilkinson and Patty, 1993 for a rationale). For the SD group, the number of chunks to be made from each given sentence was provided on the right side of the worksheet, and key words for dividing a long sentence into shorter ones were highlighted. For most sentences, the minimum chunk to be made was determined on a clause (independent and dependent) level. An example follows.

Cued SD exercises. The CEO of their company is Mrs Stark, *who* hopes that her new management strategy will boost sales. (Number of chunks: 2)

For the SC group, the authors suggested appropriate grammatical elements that could be used to combine sentences. For example, if a relative clause was required to link short sentences, a cue “**Use *who***” was given. An example is as follows.

Cued SC exercises.

- The CEO of their company is Mrs Stark.
- She hopes that her new management strategy will boost sales. **(Use *who*)**

See Appendix 3 for more examples of the SC and SD exercises.

Procedures

Based on the revised instruction techniques, the participants were given fifteen 70-min reading lessons throughout the remaining weeks, with each lesson dealing with different reading passages and a wide range of syntactic structures. In particular, the participants in the SD group were asked to decombine sentences given on an exercise sheet from a target reading passage into simple sentences, along with the aforementioned cues. The participants in the SC group were asked to combine short sentences into a longer sentence. Appropriate grammatical elements were provided for this group. Lastly, the participants in the control group were asked to read the passage and to discuss the topic of the passage with the other members of the class under the instructor’s guidance. At the end of the sixth week, a post-test was conducted to examine the effects of the treatments on the participants’ syntactic knowledge and reading comprehension.

Interviews

The first author conducted interviews at the end of the six-week program to examine how the participants in each group perceived sentence manipulation including SC and SD, and determine any unforeseen effects that were previously not experienced (e.g. reading behavior), as a result of the sentence manipulation instruction. A total of six participants volunteered for the interviews. These interviewees were from the SD and SC groups (three from each group), and they were moderate learners, based on their general class performance and performance on the tests used in the present study. With the consent of the participants, the interviews were conducted and audio-recorded by the first author. The participants' first language was used during the interviews. In each interview, the participants were first asked to describe their general experience with, and perceived value of the given activities. Subsequently, they were asked to read long English sentences aloud, to examine any unforeseen effects.

Quantitative data analysis

A set of paired *t*-tests to compare the same individuals' test scores at two different time points was conducted to identify improvements in syntactic knowledge and reading comprehension after instruction. In addition, the differences in each group's gains in terms of syntactic knowledge and reading comprehension skills were measured by analyses of covariance (ANCOVA) that controls for pre-test scores, along with Duncan's multiple-range test for possible group differences (i.e. pair-wise comparison).

In an effort to check the robustness of the findings, first the authors used one of artificial amplification techniques. Given that each group has only 15 participants, a small sample size may limit the quantitative findings; therefore, the bootstrapping technique with 100 repetitions (i.e. randomly re-sample the original cases with replacement) could be a reasonable statistical solution (Krishnamoorthy *et al.*, 2007). Second, in addition to ANCOVA, to address a potential bias that might come from differences on initial status between the experimental groups, the authors conducted the simple gain score model (i.e. analyses of variance with gain score as the dependent variable; gain score = post-test score − pre-test score), in light of Kelly and Ye's (2016) suggestion. Although the results of ANOVA for the pre-test, in terms of the homogeneity between the experimental groups, did not show any significant results, the descriptive statistics (Table I) show that the students in the control group had higher pre-test scores than their peers in the treatment groups, albeit non-significant.

Qualitative data analysis

As the interview included questions about sentence manipulation activities as well as other issues (e.g. the six-week program in which they were enrolled), only the interview data of the former were partially transcribed and translated by the first author. Due to the small amount of the interview data, the authors utilized Microsoft Word's Memo feature for the coding procedure and Microsoft Excel's spreadsheet for building the coding framework. To analyze the collected data, the authors first chose the interpretive analysis to generate explanations for how the participants perceive the provided sentence manipulation activities (Hatch, 2002). For the first cycle, in accordance with Saldaña's (2016) suggestions, the authors used descriptive coding to capture the participants' perceptions of the given activity, as well as process coding to find any differences in terms of their English reading processes related to the use of SD or SC activities. For the second cycle, the authors used pattern coding to understand relationships between and

among emerging categories to find themes that respond to the third research question. Finally, the interview data were analyzed by recurring themes. See Appendix 4 for the specific qualitative codes, categories, and themes that guided their interview analyses.

Results

Quantitative findings

Table I gives descriptive statistics for the three groups' mean scores on the pre- and post-tests, as well as inferential statistics of paired *t*-tests comparing each group's score gains before and after instruction. Also, the actual gain scores (i.e. post-test scores − pre-test scores) and their standard errors are presented in Table I. In this section, CT refers to "control", as in the control group or condition.

Overall, the results of the paired *t*-tests showed that the three groups all achieved significantly higher total scores on the post-tests, compared with the pre-tests. In particular, they performed significantly higher on the reading comprehension and syntax parts of the post-test, compared with the pre-test. However, their scores on the vocabulary section of the post-tests were not significantly different than those of the pre-tests, according to the results of paired *t*-tests.

For *RQ1*, ANCOVA was conducted to determine statistically significant differences between experimental conditions (i.e. SD, SC, or CT) on the participants' post-test scores, particularly for syntax, controlling for their pre-test scores. The results revealed that there was a significant effect of different experimental conditions on the participants' post-test syntax scores, and that this effect was large (partial $\eta^2 = 0.13$). A pair-wise comparison was further conducted to contrast differences in post-test scores across conditions (Table II). It was found that the SD group received significantly higher scores on syntax than did the SC group (p-value < 0.05). Further, these findings were still valid when the authors utilized both the bootstrapping technique with 100 repetitions and the ANOVAs with the simple gain score model. However, the comparison between the SD group and the control group in this regard was rather unclear; although the difference reaches a marginal statistical significance for two types of ANCOVAs (p-value < 0.10), it was not valid when using ANOVAs (details about this limitation are discussed in the Discussion section).

Table I.
Descriptive statistics on the pre- and post-tests

Groups	Pre-test Mean (standard deviation)				Post-test Mean (standard deviation)			
	Total	**R[a]**	**S[b]**	**V[c]**	**Total**	**R[a]**	**S[b]**	**V[c]**
SD (n = 15)	317.33 (35.45)	139 (23.46)	97.67 (11.32)	80.67 (11.93)	395.67 (39.32)	187.67 (28.09)	126.33 (11.09)	81.67 (7.72)
SC (n = 15)	317 (38.16)	146.67 (29.13)	96.33 (12.02)	74 (9.30)	363.33 (32.88)	172.33 (22.59)	114.67 (11.25)	76.33 (15.63)
CT (n = 15)	329.67 (30.96)	153 (21.53)	98.33 (15.08)	78.33 (11.90)	366.33 (28.19)	172.67 (25.56)	119.33 (11.47)	74.33 (9.23)

(continued)

Notes: **p < 0.01; ***p < 0.001; R[a]: Reading comprehension section; S[b]: Syntax section and V[c]: Vocabulary section

Table I.

Groups	*t*-test between pre- and post-tests *t* value				Gain score (post-test–pre-test) Mean (standard error)			
	Total	**R[a]**	**S[b]**	**V[c]**	**Total**	**R[a]**	**S[b]**	**V[c]**
SD (n = 15)	7.38***	5.86***	9.02***	0.30	78.33 (10.62)	48.67 (8.30)	28.67 (3.18)	1 (3.35)
SC (n = 15)	5.24***	3.91***	5.56***	0.68	46.33 (8.84)	25.67 (6.56)	18.33 (3.30)	2.33 (3.45)
CT (n = 15)	4.79***	3.12**	5.07***	−1.11	36.67 (7.65)	19.67 (6.30)	21 (4.15)	−4 (3.59)

For the second research question, on reading comprehension, an additional ANCOVA was conducted, as was done for syntax assessment. The results of the ANCOVA suggested that there was a significant large effect (partial $\eta^2 = 0.17$) of different conditions on the participants' post-test scores on the reading comprehension subsection, after controlling for their pre-test scores (see Table II). The pair-wise comparison further showed that the SD group performed significantly better than the SC group and CT group (p-value < 0.05).

Table II.
Results from between-group differences

Sections	Groups	ANCOVA (covariate: pre-test)		ANOVA (simple gain score model)	
		Contrast (SE)	Bootstrapped contrast (SE)	Contrast (SE)	Bootstrapped contrast (SE)
Syntax	SD vs SC	11.25** (3.90)	11.25** (4.01)	10.33* (5.04)	10.33* (4.94)
	SD vs CT	7.21[a] (3.90)	7.21[a] (3.90)	7.67 (5.04)	7.67 (5.18)
	SC vs CT	−4.05 (3.91)	−4.05 (4.06)	−2.67 (5.04)	−2.67 (4.02)
Reading comprehension	SD vs SC	18.50* (8.69)	18.50* (7.65)	23* (10.05)	23* (10.19)
	SD vs CT	20.78* (8.69)	20.79** (7.66)	29** (10.05)	29** (9.81)
	SC vs CT	2.29 (8.67)	2.29 (7.17)	6 (10.05)	6 (8.72)

Notes: [a]$p < 0.10$; *$p < 0.05$; **$p < 0.01$

Qualitative findings

As stated in the Methods section, interviews were conducted at the end of the six weeks to examine how the participants in the SD and SC groups perceived sentence manipulation activities they performed. In this section, SD#1 refers to the first interviewee from the SD group. Above all, SD#1 was satisfied overall with the instruction he received, as can be seen below:

> SD#1: I feel my reading comprehension skills improved through these exercises […] I gained more confidence in comprehending long and complex sentences. […] I am better at breaking down long sentences into smaller ones in my mind.

Another student, SD#3, also valued the SD activities he had performed and stated that it had enhanced his reading comprehension abilities. He described his experience with cued exercises, stating that the bold words in the long sentences from the exercises helped him recognize how each sentence was composed of several meaningful units. Furthermore, he further described how the SD activities improved his reading skills:

> SD#3: These words […] the bold ones […] tell us how to divide a long sentence into smaller ones. You can see that these smaller pieces are like units. […] After doing hundreds of SD exercises […] I can analyze the structure of the [English] sentence more quickly […] and understand it quickly.

The authors found that some students used their pens to mark places to break down sentences during reading when they were asked to read prompts aloud (i.e. long and complex sentences). This marking habit on the part of the participants in the SD group appeared to share a similar cognitive process with that emphasized by SD activities. Their markings were mostly placed at grammatically appropriate positions that correctly divided the constituents of each sentence. The students commented that they had begun to use this marking habit when they were exposed to SD activities.

It was found through these observations that some students began to use that aspect of SD as an everyday reading strategy after experiencing SD activities. Another interviewee, SD#2, took out an English fiction novel that he started to read after the program during his interview, and showed the pencil marks that he had

made in the book. The interviewer noted that the student used slashes (/) to parse sentences in accordance with their syntactic structures. This student added that he now pays more attention to the structures of complex English sentences in his comprehension of English texts than before.

On the other hand, the interviewees from the SC group described the benefits of SC instruction in terms of writing skills and highlighting specific grammatical elements, as in the following excerpt:

> SC#1: I think the exercises fostered my writing […] although they also helped me notice some grammatical aspects.
>
> SC#2: I liked the exercises […] they were useful for me to improve my English writing. […] I am less hesitant when I try to write a long sentence now, thanks to the SC exercises.
>
> SC#3: The exercise was useful […] [but] it did not seem to enhance my [syntactic] knowledge.

All interviewees from the SC group stated that the instruction enhanced their awareness of sentence structure. However, they did not mention anything related to improving their reading skills. When reading the target sentences during the interviews, the students in the SC group made a few noticeable mistakes, pausing in the middle of syntactic constituents. Overall, the responses of the three interviewees from the SC group did not allow us to detect any vestiges of their use of syntactic knowledge in the reading comprehension process.

Discussion

The present study focused on the effects of sentence manipulation activities on the development of reading comprehension among L2 adult learners during a six-week English language learning program. Overall, in comparison with the pre-test, all three groups scored higher on both the syntax section and the reading comprehension section on the post-test. However, it was found that SD was more effective in enhancing syntactic knowledge and reading comprehension than the other conditions, although the statistical significance for the comparison between the SD group and the control group on the syntactic part was marginal. Taken together, the finding suggests that SD activities, at least for the participants of this study, made a skill transfer possible (i.e. from writing to reading), which may be mediated by syntactic knowledge (Fitzgerald and Shanahan, 2000).

Overall, the findings revealed that SD was more beneficial to the participants than the other conditions. The SC and control groups' scores on the set of questions on syntactic knowledge increased significantly after the program; however, the degree of increase was much larger for the SD group. This finding accords with the prediction of Strong (1986), who suggested that decombining exercises may be more useful for those who are developmentally ready in terms of psycholinguistic stage. Although Strong did not elaborate on his description of this stage, the authors can postulate that one must be equipped with a certain level of conscious understanding of how syntactic structures work on a sentential level to significantly benefit from this type of cognitively and linguistically challenging instruction. That being said, learners with a low level of cognitive maturity (e.g. young children) and at a beginning level of English proficiency, who may still grapple with lower-level English language processing skills (Bernhardt, 1991), may not be ideal beneficiaries of sentence manipulation instruction. Yet, as there is a lack of previous research on this issue, there is a need for more studies that examine this hypothesis thoroughly.

Importantly, this study found that the higher performance on the reading comprehension questions on the part of SD group may have resulted from an increase in syntactic knowledge. Thus, a theoretical link among SD, syntactic knowledge, and reading comprehension can be suggested based on the design of the present study, in which the only difference among the groups was sentence manipulation exercises (see Nation and Snowling, 2000; Shiotsu and Weir, 2007 for more discussion on a theoretical link between syntactic knowledge and reading comprehension). Also, the finding neatly fits in with Bernhardt's (1991) model of L2 reading, in that the increase in syntactic knowledge for intermediate-level L2 learners would lead to an enhancement in reading comprehension.

The results of the interviews with the participants generally corresponded to the greater effectiveness of SD over SC in terms of enhancing reading comprehension from the results of the post-tests. For example, students in the SD group perceived SD activities as enhancing their syntactic processing, noting the merits of that type of instruction in terms of enabling them to break down syntactically complex sentences into meaningful chunks, which in turn appears to have enhanced their reading comprehension.

A rather different pattern was observed among students who performed SC activities. All of the students in the SC group described benefits of SC activities in terms of boosting their confidence when composing syntactically complex and long sentences. Hence, SC was perceived as writing-oriented by the participants, in accordance with its original purpose (Hunt, 1965; Mellon, 1969). Although the interviewees also noted that SC activities enhanced their awareness of particular syntactic structures, the results from the experiment and the interviewees' comments suggested that this increase in self-perceived awareness did not necessarily lead to the development of reading comprehension. Overall, the substantial differences in perception toward sentence manipulation activities between students who experienced SD and SC lend credence to the authors' hypothesis that SD was more effective than SC in enhancing reading comprehension.

The present study had some limitations. First, the small sample size of the present study would limit the statistical findings of the present study, though the authors adopted a few statistical adjustments and approaches to overcome the potential bias that might come from a seemingly difference in initial status (pre-test scores) between the control and experimental groups. The authors also believe that this may have caused them to encounter an inferential challenge for the comparison of the test scores on the syntactic knowledge section between the SD and control group. Second, the long-term effects of sentence manipulation instruction could not be evaluated due to practical constraints. Although the findings of the present study favored SD activities, whether SC activities bring about similar long term learning outcomes among L2 learners, or are more effective than SD in terms of other aspects of English learning remain unclear.

Conclusion

The present study examined the effects of SD and SC on reading comprehension among L2 learners. In short, we found that, for those L2 learners who have passed a pre-intermediate threshold level, SD is preferable to SC for enhancing syntactic knowledge, which, in turn, appears to contribute to better reading comprehension. The finding of the present study has pedagogical implications for ELA classrooms in English-speaking countries (Australia, the UK, the USA, etc.) where both NESs and ELLs are learning English literacy. Given that previous studies in this context have shown that SC activities bring about a positive effect on improving students' reading and writing skills, particularly for young learners who are developing their syntactic knowledge, we can further assume that SD activities could have some positive

effects on these (or other) aspects of academic literacy, which calls for future research. In addition, the finding of the present study highlights the value of using pedagogical practices aimed at enhancing linguistic awareness through writing exercises for L2 learners. In this way, ELA teachers, who might not be familiar with this teaching technique, can study and examine syntactic constituents of the English language, particularly from the perspective of ELLs. Thus, formal training in sentence manipulation instruction should be mandatory for ELA teachers, so that they can be instructed on how to adapt existing sentence manipulating exercises to their own teaching contexts, in terms of the characteristics of their student population. Overall, educators and teachers are encouraged to exploit sentence manipulation activities (i.e. SD and SC) as an effective English teaching strategy according to the learner population.

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Appendix 1

Table AI.
Sample question items in the reading sections of the pre- and post-tests (Adapted from Educational Testing Service, 2015)

Question type	Sample question
Question on vocabulary	102. The award is given to an individual who has made _____ contributions to the community through volunteer work (A) detailed (B) significant (C) secure (D) updated

Question type	Sample question
Question on syntactic (grammatical) knowledge	101. A late fee will be applied to your account _____ payment is not received by March 31 (A) and (B) whether (C) but (D) if
Question on reading comprehension	Questions 153 refers to a notice about a public sale (For the detail see Educational Testing Service, 2015) 153. What is the purpose of the notice? (A) To announce a sale of artwork (B) To advertise the opening of a hotel (C) To offer a discount on painting lessons (D) To publicize a photography exhibition

Note: ETS TOEIC® Examinee Handbook Listening and Reading – Sample Questions 101, 102 and 153 Copyright © 2015 Educational Testing Service are reprinted by permission of Educational Testing Service

Appendix 2

Table AII.
Target syntactic elements

Syntactic structure	Subcategorization	Grammatical cues to be used in de-combining or combining sentences in SD and SC exercises
Adverbial clause	condition	if, unless
	purpose	in order to, so that
	concession	although, though, while
	time	before, after, since, while, as, by, until
	reason	because, since, as, owing to
	place	where, wherever
Relative clause	restrictive	who, which, that
	nonrestrictive	who, which
Non-finite clause		for/to
Coordinating conjunction		and, but, yet, so, or, for
Participial phrase		-ing endings
Appositive phrase		comma

Appendix 3. Examples of SC and SD exercises

Cued SC exercises

(1) a. It was the first time.

b. The brand new MP3 player hit the stores yesterday. **(Use *that*)**

(2) a. The customers arrived late to our office.

b. There was very little time to talk about a new deal. **(Use *because*)**

(3) a. I purchased the house located in the city center.

b. You really liked the house located in the city center. **(Use *that*)**

c. I hope your family will visit us often. **(Use *and*)**

Cued SD exercises

(1) It was the first time *that* the brand new MP3 player hit the stores yesterday. **(2)**

(2) *Because* the customers arrived late to our office, there was very little time to talk about a new deal. **(2)**

(3) I purchased the house *that* you really liked located in the city center, *and* I hope your family will visit us often. **(3)**

Appendix 4

Table AIII.
Qualitative data matrix for the interview analyses

Selected codes and examples	Selected categories	Themes
Usefulness ("SC#3: The exercise was useful.")	SD and SC learners have positive attitudes towards the activities	Learners' general perception towards and interests in sentence manipulation activities
Interests (SD#3: "I really enjoyed the activities.")	SD and SC learners have interests in the activities	
Improvement ("SD#1: I feel my reading comprehension skills improved through these exercises.") Confidence ("SD#1: I gained more confidence in comprehending long and complex sentences.")	SD learners think the activities enhanced their reading comprehension skills	Learners' perceived pedagogical value of sentence manipulation activities
Grammar improvement ("SC#1: they also helped me notice some grammatical aspects.") Writing improvement ("SC#2: they were useful for me to improve my English writing.")	SC learners think the activities helped them improve grammatical aspects of writing	
Using slashes (/) to parse sentences (SD#2) Paying more attention to the structures of complex English sentences (SD#1, 2, and 3) Pausing mistakes while reading (#SC 1, 2, and 3)	SD learners exhibit their gained syntactic awareness	Learners' increased awareness of English sentence structures
Understanding English sentence structures while reading ("SD#1: I am better at breaking down long sentences into smaller ones")	SD and SC learners think they gained syntactic awareness	

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